

Influence of a large fluvial island, streambed, and stream bank on surface water-groundwater fluxes and water table dynamics

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[1] Substantial research on how hydraulic and geomorphologic factors control hyporheic exchange has resulted in reasonable process understanding; however, the role of fluvial islands on the transient nature of spatial flux patterns remains elusive. We used detailed field observations of the Truckee River, Nevada from 2003 to 2009 to quantify fluid flux between the river and a fluvial island, the streambed, and the adjacent stream bank. We constructed a 3-D numerical flow and heat transport model to further quantify the complex flow paths. Our study expands on previous research typically confined to less comprehensive scales and dimensions, and highlights the transient multidimensionality of the flow field. In fact, 1-D vertical streambed flux estimates indicated that the channel bar tail displayed the highest upward flux throughout the summer; however, 3-D model results indicated that the horizontal contribution was two orders of magnitude higher than the vertical contribution. The channel bar net flux is typically 1.5 orders of magnitude greater than the adjacent stream banks and an order of magnitude less than net streambed fluxes, indicating significant differences in river-aquifer interactions between each of the geomorphic units. Modeling simulations further indicated that the channel bar induces 6 times more fluid flux than an identical location without a fluvial island, consistent with flux estimates from a nearby river restoration location. Moreover, event-based and seasonal transient antecedent moisture and near-stream storage conditions contribute to multidimensional river-groundwater interactions. These results suggest that fluvial islands are a key driver and significant component of river-groundwater interactions and hyporheic flow.

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1. Introduction

[2] Channel bars are a common in-stream geomorphic feature in a large range of natural and regulated river classes, and their impact on stream water interactions with shallow groundwater has only recently been examined [Wyrick and Klingeman, 2011]. For this study, a channel bar is defined as a large, midriver fluvial island that is entirely separated from the floodplain by water, exhibits stability, is inundated during bankfull conditions, and has established permanent vegetation [Osterkamp, 1998; Wyrick and Klingeman, 2011]. Channel bars induce river-groundwater interactions by chang-

ing flow patterns which affect biogeochemical reactions, stream water quality, and near-stream ecological habitats [Dent et al., 2001; Fanelli and Lautz, 2008]. While the geologic structure of channel bars have previously been described [i.e., Ashworth, 1996], fluid flow and transport patterns through channel bars have largely been overlooked.

[3] Several classical studies have examined the importance of near-stream geomorphic features on flow patterns, temperature adjustment, and transport. Wondzell and Swanson [1999] found that while lateral flow typically dominates through gravel bars, a large flood event created extensive hyporheic zones with increased downward flux at the head of several gravel bars while lateral downstream river-bar interactions were maintained. In fact, daily dam-induced stage changes were found to cause rapid and extensive lateral head variations throughout a large Texas channel bar [Francis et al., 2010]. Alternatively, Wroblicky et al. [1998] found that increased stream discharge during peak flow conditions decreased the lateral river-aquifer interaction distance by as much as 50%. Flow paths through channel bar deposits into adjacent alcove surface water were found to modify channel bar water quality and decrease temperature, particularly with highly permeable substrate material [Arrigoni et al., 2008; Fernald et al., 2006]. Both natural and constructed gravel bars have also been shown to induce

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nutrient removal [Harvey and Bencala, 1993; Kasahara and Wondzell, 2003; Kasahara and Hill, 2007]. Further, several studies have found that only near-stream gravel bar edges increased surface water nutrient concentrations [Dent et al., 2001; Zarnetske et al., 2011]. Fernald et al. [2006] suggest that the near-stream interactions can be enhanced by restoring in-stream processes that contribute to gravel bar deposits.

[4] Flow paths, driven by morphologically induced head gradients, range from millimeters to hundreds of meters with residence times from seconds to years [Harvey and Bencala, 1993; Thibodeaux and Boyle, 1987]. Pore-scale subsurface flow paths are nested within spatially and temporally increasing regional flow paths that leave the river and return to the channel within the reach of interest [Harvey and Bencala, 1993; Harvey and Wagner, 2000]. This overall river-groundwater interaction is generally termed hyporheic exchange and includes vertical streambed recharge and discharge, lateral bank storage, and longitudinal streambed exchange. Local-scale streambed topography induces pressure variations causing flux into and out of the streambed [Elliott and Brooks, 1997a, 1997b; Thibodeaux and Boyle, 1987]. These river-groundwater interactions are influenced by hydraulic conductivity magnitude and distribution [Cardenas and Zlotnik, 2003] and in-stream velocity distributions [Mendoza and Zhou, 1992]. On a moderate-scale, in-stream obstructions [Wondzell et al., 2009] and riffle-pool sequences [Gooseff et al., 2006; Woessner, 2000] further induce river-aquifer interactions. On the reach scale, heterogeneity and anisotropy [Fleckenstein et al., 2006; Niswonger and Fogg, 2008], variations in stream curvature [Revelli et al., 2008], stage [Barlow and Coupe, 2009], and stream position within the fluvial plain [Woessner, 2000] influence larger flow paths.

[5] Kalbus et al. [2006] discussed several approaches to quantify river-groundwater interactions, including using heat as a natural tracer. Heat as a tracer is an effective method for estimating river-groundwater fluxes primarily because sediment thermal conductivity varies little relative to the hydraulic conductivity in streambeds and is independent of sediment texture [Cardenas and Zlotnik, 2003; Constantz, 2008]. Temperature measurements have successfully been used to estimate river interactions with shallow sediments at sites ranging from small mountain streams [Conant, 2004; Fanelli and Lautz, 2008; Swanson and Cardenas, 2010] to large, gravel bed rivers [Burkholder et al., 2008]. River-groundwater interactions are quantified by measuring temperature at a range of depths below the streambed and then calculating temperature attenuation with depth. Infiltration rates are estimated with the convection-conduction equation by varying the hydraulic conductivity. One-dimensional (1-D) analytical solutions [i.e., Stallman, 1965] have successfully been used to examine vertical flux rates. Increasingly, robust numerical models have been constructed to simulate 2-D and 3-D near-stream flux rates [Kasahara and Wondzell, 2003; Storey et al., 2003]. The theory, development, and application of heat as a tracer is thoroughly discussed by Stonestrom and Constantz [2003], Anderson [2005], and Constantz [2008].

[6] Though channel bars are ubiquitous throughout many river systems, their role in near-stream river-groundwater interactions is poorly understood. The cumulative objective of this study is to compare the relative magnitude

river-channel bar fluxes with the fluxes at the adjacent river-streambed and the river-stream bank. We use a long-term field-based data set from 2003 to 2009 in order to examine the spatial flux variability between a fluvial island, the streambed, and the adjacent stream bank throughout the study area. To further investigate the multi-dimensional spatial flux contributions, we analyze a winter subset of the data set to construct a 3-D flow and heat transport model. We also investigate the transient dynamics of the field-based data set to show how antecedent moisture conditions and aquifer storage dynamics from previous events control near-stream fluxes and both spatial and temporal flow patterns should be taken into consideration.

2. Study Site

[7] The study area is located 27 km east of Reno, Nevada, in the lower Truckee River watershed on the eastern slope of the Sierra Nevada (Figure 1). The 191-km long Truckee River begins at Lake Tahoe in northern California and terminates at Pyramid Lake in northwestern Nevada. The Truckee River flows from a high mountain to arid ecosystem, creating cascading meteorological forcing conditions. Mean annual precipitation along the river corridor ranges from 660 to 1220 mm yr⁻¹ in the upper Lake Tahoe region to less than 170 mm yr⁻¹ near the terminus of the river [Shope, 2009]. Annual precipitation for Reno, Nevada is less than 178 mm yr⁻¹, and the mean annual air temperature near the study area is 12.4°C. Mean annual river discharge under base flow conditions at the United States Geological Survey (USGS) Tracy, Nevada streamgauge (10350340), located 1.1 km downstream from the study area, is 12.9 m³ s⁻¹. The hydrograph is dominated by snowmelt occurring around May and slowly decreasing after this time [Shope, 2009].

[8] The area of interest is an unnamed representative in-stream channel bar located on the McCarran Ranch property and currently owned and operated by The Nature Conservancy. The channel bar is therefore referred to as the McCarran Ranch Channel Bar (MRCB). The study reach and MRCB (Figure 1) consist of mixed grain size alluvial and colluvial deposits [Shope, 2009]. The gravel-bedded sixth-order Truckee River has an average bed slope of 0.185% within the study reach [Shope, 2009]. Under base flow conditions, the MRCB is 198 m long, 60 m wide, averaging 1302 m above sea level (asl) and a maximum of 1.2 m above the river stage. The river stage is 0.2 m higher on the east bank than the west bank during base flow conditions. The MRCB is inundated on average every 2.5 yr at a surface water flow rate of 141 m³ s⁻¹ [Shope, 2009]. Approximately 30% of the lateral MRCB edge is covered with Fremont Cottonwood (*Populus fremontii*), willow (*Salix boothii*), and Mountain Alder (*Alnus incana*), while the remainder of the bar is bare soil (Figure 1). The Truckee River is generally losing to the aquifer in this area [Sada et al., 2005], although field data identified local gaining conditions, facilitated by stream geometry and streambed geomorphology.

3. Methods

3.1. Field Measurements

[9] A 300 m reach of the Truckee River encompassing the MRCB was chosen for investigation based on previous

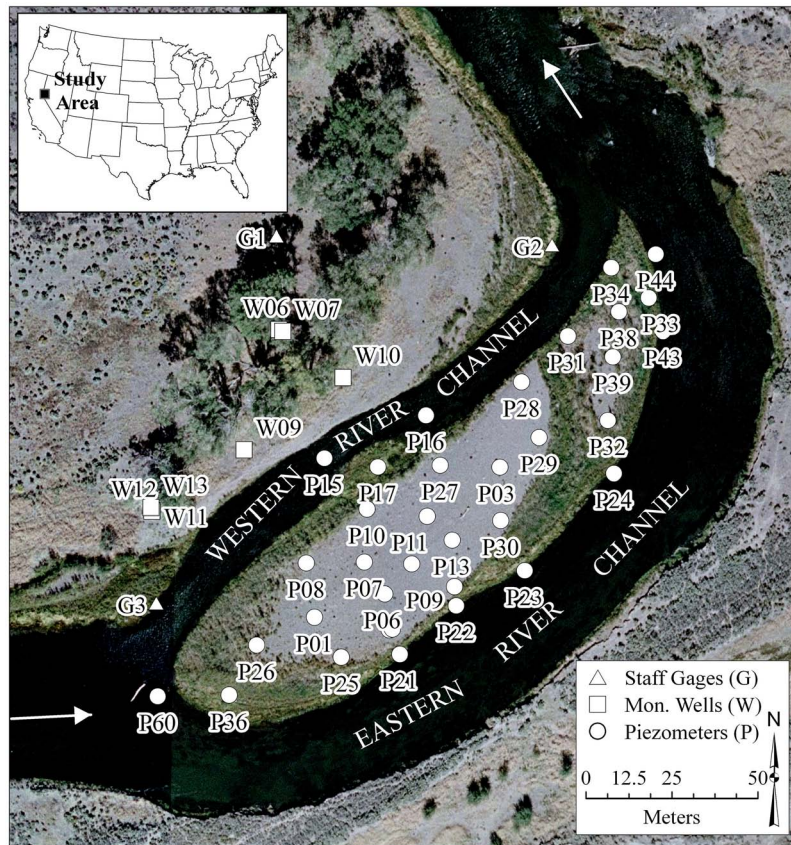


Figure 1. The study location is 27 km east of Reno, NV (39.5469, -119.5585). Piezometer, monitoring well, and staff gauge locations throughout the channel bar, the streambed, and the stream bank are shown and streamflow is from lower left to upper right. The eastern and western river reaches adjacent to the McCarran Ranch Channel Bar (MRCB) are annotated.

studies indicating highly variable river-groundwater interactions [Sada *et al.*, 2005]. Understanding flow patterns and surface water-groundwater interactions between the river system and the channel bar, the streambed, and the stream bank requires information on transient precipitation, temperature, and river stage. The extended asymmetric shape of the MRCB warranted a unique pattern of monitoring points; so, the channel bar and streambed were instrumented with a dense, horizontally and vertically distributed array of 58 shallow PVC piezometers (Figure 1). The piezometers were constructed of 2.1-cm diameter PVC casing to depths ranging between 1.11 to 1.78 m below ground surface (bgs) and screened along the bottom 2–5 cm. The adjacent stream bank ground surface elevation was slightly higher and required deeper wells constructed of 5.1-cm diameter PVC casing to depths of between 3.91 to 10.69 m bgs and screened along the bottom 50 cm.

[10] The hydraulic head in wells, piezometers, and the river were monitored between 2003 and 2009 with Micro-Diver absolute pressure transducers (Schlumberger Water Services, Inc.) and water level recorders (Tru-Track, Inc.). Continuous head measurements were collected at 30- or 60-minute intervals and verified at least monthly with manual measurements (± 0.002 m). Twenty-two variable head hydraulic conductivity tests were conducted at 6 channel bar piezometers (Table 1). The piezometers, the topography, and

the river channel bathymetry were periodically surveyed using a transit survey system (Nikon DTM 450, ± 0.002 m).

[11] Temperature was monitored between 2003 and 2009 in all wells, piezometers, and the river with single-channel, thermistor dataloggers (Maxim iButton, model DS1921Z). Prior to and after deployment, all dataloggers were calibrated in a four-point water bath encompassing the full range of expected temperatures and a correction factor was established for individual dataloggers. Following this procedure, the calibrated temperature measurements were accurate to within $\pm 0.2^\circ\text{C}$. A vertical array of three to 30 temperature dataloggers were deployed in each piezometer with measurements recorded at 60-minute intervals. Stream

Table 1. Falling-Head Slug Test Hydraulic Conductivity Estimates (m s^{-1}) for Locations Throughout the Channel Bar

Location	Hydraulic Conductivity (m s^{-1})	Range (m s^{-1})	Repetitions
P01	$3.29\text{E-}05 \pm 4.42\text{E-}06$	$2.81\text{E-}05$ to $3.84\text{E-}05$	5
P02	$5.43\text{E-}05 \pm 4.52\text{E-}06$	$5.12\text{E-}05$ to $6.09\text{E-}05$	4
P03	$2.89\text{E-}07 \pm 5.64\text{E-}08$	$2.49\text{E-}07$ to $3.29\text{E-}07$	2
P05	$2.17\text{E-}04 \pm 1.48\text{E-}05$	$2.03\text{E-}04$ to $2.33\text{E-}04$	3
P07	$2.16\text{E-}05 \pm 2.32\text{E-}06$	$1.98\text{E-}05$ to $2.49\text{E-}05$	4
P30	$5.43\text{E-}05 \pm 4.52\text{E-}06$	$5.12\text{E-}05$ to $6.09\text{E-}05$	4

temperature was also measured at each in-stream piezometer within 0.05 m of the streambed.

[12] River discharge was periodically measured at 28 locations between 2003 and 2009 throughout the 6 km reach encompassing the study site, including two locations immediately above and below the channel bar [Sada *et al.*, 2005; Shope, 2009]. River discharge was estimated by multiple methods including the velocity-area technique [Rantz, 1982] with an electromagnetic velocity sensor (Marsh-McBirney Flo-Mate, Hach Co.) and an Acoustic Doppler Current Profiler (ADCP) (Sontek/YSI, Inc.) The USGS Tracy, Nevada stream gauge (10350340), located 1.1 km downstream from the channel bar study area provided representative 15-minute, real-time stage and in-streamflow rates. Multiple river discharge measurements collected immediately upstream and downstream from the MRCB were compared to the USGS gauge discharge and differences were found to be negligible.

[13] We calculated the seepage flux or specific discharge, q (m s^{-1}), with Darcy's law, $q = -K(\Delta h/\Delta z)$ where, K is the saturated hydraulic conductivity (m s^{-1}), and $\Delta h/\Delta z$ is the linearly varying hydraulic gradient (m m^{-1}). The vertical streambed flux was typically estimated through the streambed boundary. At stream bank locations, the vertical flux was estimated through nested wells. As expected in fluvial systems, field observations indicate significant near-stream layering and event-based aggradation/degradation sequences. Depositional processes can cause anisotropic flow conditions up to two orders of magnitude. Therefore, an anisotropy ratio of 4 was assumed, similar to Chen [2004], resulting in an average vertical hydraulic conductivity estimate of $2.2 \times 10^{-6} \text{ m s}^{-1}$. The spatially distributed slug test hydraulic conductivity values were interpolated throughout the MRCB based on soil texture and sediment stratigraphy. Horizontal hydraulic gradients were interpolated from the previously described transient head distributions and coupled with hydraulic conductivity to estimate horizontal seepage fluxes.

3.2. Groundwater Flow Modeling

[14] To corroborate and support spatial field-based flux patterns between the river and the MRCB, the streambed, and the stream bank, a 3-D flow and heat transport model was constructed for a short, highly instrumented period from 9–14 February 2008. The widely used, USGS MODFLOW code was used to solve the steady state groundwater flow equation [McDonald and Harbaugh, 1988]. The model domain extended $553 \text{ m} \times 550 \text{ m}$ with a $15.7 \text{ m} \times 15.8 \text{ m}$ horizontal grid discretization and a 10 layer vertical resolution, sufficient to capture overall flow paths. The top of the first layer was a composite of the field-surveyed digital elevation model (DEM) and the streambed bathymetry. The second layer upper elevation was 0.1 m below the streambed and the adjacent potentiometric surface. The next six model layers were uniformly subtracted in 0.2 m intervals to a depth of 1.3 m below the water table/streambed, which enabled resolution of shallow flow paths. The ninth layer was of variable thickness to a bottom elevation of 1294 m. The bottom layer (10) was prescribed a 2 m uniformly thickness. The top eight model layers represented the shallow subsurface and were prescribed three discrete hydraulic conductivity values of 2.3×10^{-5} , 3.0×10^{-4} , and $2.0 \times 10^{-6} \text{ m s}^{-1}$ for the channel bar, the streambed,

and the stream bank, respectively (Table 1). The bottom two layers were assigned a hydraulic conductivity of $2.3 \times 10^{-5} \text{ m s}^{-1}$, representing the geometric mean of all slug test measurements.

[15] The simulated period was chosen because the river stage was relatively constant ($\pm 0.015 \text{ m}$) and therefore, the head distribution was considered steady state. Spatially variable constant head stream boundaries were linearly interpolated between field-based surveyed elevations and prescribed to stream channel cells (Figure 2). Precipitation was insignificant for the midwinter simulation period and the preceding 10 d. Total potential evapotranspiration (ET) measured in February was $<5\%$ of midsummer values at a similar location 12 km away. Stream bank well (W06) had a nearly linear temperature profile with depth, suggesting low vertical flux and further supporting the assumption of limited water movement due to precipitation and ET [Shope, 2009]. Therefore, the MRCB and stream bank were prescribed no-flow boundary conditions (Figure 2).

[16] Lateral boundaries to the west, north, and east were prescribed spatially variable constant-head boundaries from the regional McCarran Ranch potentiometric surface [Sada *et al.*, 2005]. At the southern boundary, the stream terrace and rock outcrop were assigned a no-flow boundary condition. The bottom model boundary represents regional downgradient flow and low hydraulic conductivity relative to the near-stream alluvium and was assigned a no-flow condition. Hydraulic properties are summarized in Table 2 and boundary conditions in Figure 2.

3.3. Heat Transport Modeling

[17] Heat transport in porous media is expressed by the convection-dispersion equation which is analogous to the advection-dispersion equation for solute transport [Constantz, 2008]. A form of the convection-dispersion equation can be written as [Anderson, 2005; Ingebritsen and Sanford, 1998],

$$\frac{\partial T}{\partial t} = K_e \nabla^2 T - \frac{\rho_w c_w}{\rho c} \nabla \cdot (qT), \quad (1)$$

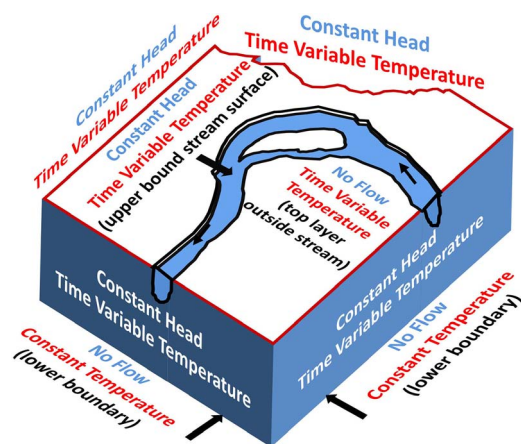


Figure 2. Conceptual fluid flow and temperature boundary conditions for the 3-D simulated domain. Fluid flow boundary conditions are presented in blue and temperature boundary conditions in red. Streamflow direction is indicated by arrows.

Table 2. Summary of Hydraulic Parameters, Thermal Parameters, and Boundary Conditions Used for Unsaturated Zone Infiltration and Multidimensional Modeling Simulations

Parameter	Description	Value	Unit
λ_s	Saturated effective thermal conductivity	1.8 ^a	(W m ⁻¹ °C ⁻¹)
λ_u	Unsaturated effective thermal conductivity	0.245 ^b	(W m ⁻¹ °C ⁻¹)
ρ_w	Water density	1000	(kg m ⁻³)
ρ_s	Saturated sediment density	1800 ^c	(kg m ⁻³)
ρ_u	Unsaturated sediment density	1310 ^c	(kg m ⁻³)
ρ_B	Saturated sediment bulk density	1800 ^c	(kg m ⁻³)
c_w	Specific heat of water	4180 ^d	(J kg ⁻¹ °C ⁻¹)
c_s	saturated sediment specific heat	1910 ^c	(J kg ⁻¹ °C ⁻¹)
c_u	Unsaturated sediment specific heat	782 ^b	(J kg ⁻¹ °C ⁻¹)
K_e	Effective thermal diffusivity	0.045 ^d	(m ² d ⁻¹)
n (sand)	Porosity	0.4	
K	Hydraulic conductivity	2.2×10^{-5}	(m s ⁻¹)
α	Van Genuchten alpha	0.145 ^f	(cm ⁻¹)
n_{VG}	Van Genuchten n	2.68 ^f	
Θ_s	Saturated water content	0.4	
Θ_r	Residual water content	0.045 ^f	
m_{VG}	Van Genuchten m	0.627	
x	Maximum channel bar length and width	198×59	(m)
S	Average bed gradient	0.185	(m m ⁻¹)
w	Channel width	Variable	(m)
T_{air}	Upper boundary	Variable	(°C)
T_{GW}	Lower boundary	13.2	(°C)

^aMidpoint of range from Niswonger and Prudic [2003].^bde Vries [1966]; van Wijk [1966]; Hanks [1992].^cLapham [1989].^dJury et al. [1991].^eConstantz [2003].^fCarsel and Parrish [1988].

where T is temperature, ρ and ρ_w are the sediment-fluid bulk density and water density, c_s and c_w are the specific heat capacities of the sediment-fluid and the water, q is the seepage velocity or specific discharge vector, and K_e is the effective thermal diffusivity. The effective thermal diffusivity represents both the transport of heat by sediment-fluid conduction and thermal dispersion and is expressed as [Hatch et al., 2006],

$$K_e = \frac{\lambda}{\rho c} + \beta |q|, \quad (2)$$

where λ is the bulk thermal conductivity of the fluid-sediment mix and β is thermal dispersivity.

[18] To quantify heat transport and simulate river and groundwater interactions, MT3D was used to solve the advection-dispersion transport equation after MODFLOW solves the groundwater flow equation [Zheng and Wang, 1999]. Effects of temperature derived buoyancy and viscosity were decoupled from flow and neglected in our simulations, as supported by Ma and Zheng [2010]. Conduction in MT3D was accounted for by parameterizing the effective molecular diffusion coefficient equal to K_e (Table 2). In experimental thermal dispersion investigations, Rau et al. [2012] indicated that in relatively coarse sand, thermal dispersivity can be neglected if the thermal Peclet number is <0.5 , similar to our study site. The experimental velocities were also nearly 30 times greater than our estimated velocities. At experimental velocities similar to our field estimates,

there was significant scatter in longitudinal effective dispersion. The experimental scatter was suspected to result from heterogeneity, which would exhibit greater influence in our field site than in a laboratory setting. Other investigations have found thermal dispersivity to have a minimal effect on residence time [i.e., Bear, 1972; Elliott and Brooks, 1997a, 1997b], and estimates have been inconsistent with disputed and controversial parameter values [i.e., de Marsily, 1986; Ingebritsen and Sanford, 1998; Niswonger and Prudic, 2003]. We simulated the effects of thermal dispersion by prescribing β between 0.01 and 1 m [Niswonger and Prudic, 2003]. However, the results were typically negligible and ultimately neglected in our final simulations.

[19] Variably saturated heat transport models have also been used to examine river-groundwater interactions [i.e., Burrow et al., 2005]. However, multidimensional, variably saturated flow and energy transport is computationally expensive and the increased precision in river-groundwater flux estimation may be unnecessary for the observed conditions. As discussed, limited vertical water movement due to precipitation and ET was assumed and verified by a deep study area well [Shope, 2009]. Further, the focus of this study was to examine interactions between the river and the saturated aquifer, and therefore the fully saturated MODFLOW/MT3D simulation was sufficient.

[20] To prescribe the upper boundary temperature conditions, two steps were required. First, the ground surface skin temperature (GST) was estimated from the measured air temperature, and second, the water table temperature was computed from the GST. Air temperature cannot be used as a proxy for GST because it does not account for net radiation, soil heat flux, latent heat flux, sensible heat, albedo, vegetative cover, conductance, or soil moisture content. Hourly air temperature and GST from seven locations with similar characteristics to the study area [Shope, 2009] were used to develop a binary regression. Hourly GST was, on average, 1.15 ± 0.10 (day) and 0.96 ± 0.19 (night) times the air temperature and used to calculate the study area GST from the measured air temperature. To estimate the water table temperature from the GST, a 1-D vertical, finite difference solution to the heat transport equation (equation (1)) was used with hourly GST as the upper boundary. The soil water content-pressure head model [Van Genuchten, 1980] was used with the statistically derived water retention parameters of Carsel and Parrish [1988] to estimate the variably saturated moisture content. Unsaturated soil, thermal, and hydraulic properties are summarized in Table 2. A 45 d ramp-up period was provided for the initial conditions. The MRCB and stream bank upper-temperature boundary was spatially and temporally variable while the bottom boundary was representative of constant groundwater temperature at 13.2°C. Because in-stream temperature varied $<0.1^\circ\text{C}$ in the water column throughout the length of the study reach, stream cells were prescribed spatially constant, transient temperatures. Cells at the west, north, east, and south boundaries were prescribed linearly interpolated temperature between the top and bottom of the model domain. Refer to Figure 2 for all temperature boundary conditions.

3.4. Inverse Estimation of Hydraulic Conductivity

[21] The simulated versus observed head and temperature residuals throughout the modeled period were

minimized through several objective functions by varying the hydraulic conductivity, as described by *Niswonger and Prudic* [2003]. To automate the procedure, the nonlinear parameter estimation code PEST [Doherty, 2004] was used to calibrate hydraulic conductivity values. To evaluate the modeling performance, several criteria were used including: the coefficient of determination (R^2), the Nash-Sutcliffe coefficient of efficiency (NSE), the root-mean-square error (RMSE), and mean absolute error (MAE) [Helsel and Hirsch, 1992; Legates and McCabe, 1999; Nash and Sutcliffe, 1970]. R^2 describes the percentage of variability explained by the model and ranges from 0 to 1, with higher values indicating less error variance. NSE ranges between $-\infty$ and 1.0 where a NSE of 1 is the optimal value. NSE may be a better predictor than R^2 ; although, NSE and R^2 are both sensitive to extreme values because they square the paired differences. MAE describes the average of the absolute error between the simulated and observed values and often is confused with the mean absolute deviation, which describes the average difference between an individual value and the mean of the entire data set. Both the MAE and RMSE range between 0 to ∞ and are negatively oriented, where values approaching zero indicate greater agreement between residuals.

4. Results and Interpretation

4.1. Vertical Fluid Flux Estimates From Field Data

[22] As expected, vertical streambed flux direction and magnitude was spatially variable and correlated to near-stream geomorphic conditions. Representative midsummer vertical flux distribution at eight of the 11 streambed locations is presented in Figure 3; however, three locations are described in further detail. The upstream location (P60) is the only flux distribution with an entirely downward direction; during the summer, the MRCB tail location (P44) is consistently higher than other locations; and location P17, adjacent to the MRCB, is predominately upward flux with up to diel direction fluctuations.

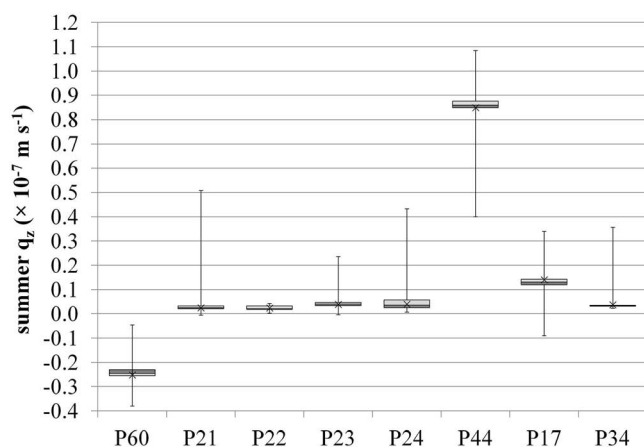


Figure 3. Representative 1-D vertical streambed flux (q_z) distribution for eight of the 11 streambed locations calculated from summer 2007. The “x” represents the average flux distribution. Locations P60, P44, and P17 are described in further detail in Figure 4.

[23] The vertical hydraulic gradient (VHG), vertical flux, river discharge, and vertically distributed temperatures at locations P60, P17, and P44 are presented in Figure 4, for representative summer and winter periods. The total measured streambed flux ranged between -0.56 and $+1.23 \times 10^{-7} \text{ m s}^{-1}$ with the highest magnitude flux in the upward direction at the MRCB tail (P44) in late summer. As a comparison, the summer vertical flux in the stream bank ranged between -1.56 and $+2.04 \times 10^{-8} \text{ m s}^{-1}$ [Shope, 2009]. A summary of the seasonal VHG and vertical seepage flux range at each of the three streambed locations is provided in Table 3. The highest flux magnitude at the MRCB head (P60) was estimated during winter conditions and was 5–10 times higher than the autumn or spring conditions when seepage typically reduced to near-neutral conditions during peak events. Alternatively, the highest seepage at the MRCB tail (P44) was predominately upward through the late spring and summer. During the winter, near neutral flux was observed at the tail with significant variability in both magnitude and direction (Table 3). Seepage at the tail generally changed direction during the winter and spring. At location P17, the average flux direction was upward with the highest magnitude in the spring, although the flux direction varied with events in the summer. Near-neutral seepage during the autumn and winter periods coupled with diurnal stage variations triggered diurnal flux direction changes at P17 [Shope, 2009]. A nonlinear stage height threshold is observed where the flux direction changes, particularly at P17.

[24] The Darcy-derived vertical river-streambed fluxes (Figure 4) are reasonable when compared with temperature-derived vertical seepage estimates from equation (1) [Shope, 2009]. The temperature-derived flux was optimized over an entire 78-d winter period and the MRCB head and tail (P60 and P44) have the same direction and similar magnitude as the Darcy calculated fluxes (Table 3). While the hydraulic vertical river-streambed fluxes were calculated with a constant vertical hydraulic conductivity of $2.2 \times 10^{-6} \text{ m s}^{-1}$ estimated from slug tests, the inversely optimized heat-based vertical hydraulic conductivity values for P60, P17, and P44 were $1.3 \times 10^{-6} \text{ m s}^{-1}$, 3.3×10^{-6} , and $4.9 \times 10^{-6} \text{ m s}^{-1}$, respectively.

[25] The streambed thermographs for locations P60, P17, and P44 (Figure 4) provide clear evidence of diel temperature changes that decrease in magnitude and are phase-shifted forward with depth, further elucidating the vertical flow direction. In general, streambed thermographs correlate with the river discharge and estimates of vertical flux direction (Figure 4). Greater temperature amplitude is observed at -0.207 m depth bgs in P60 (Figure 4A) than at similar depths in P17 (Figure 4B) or P44 (Figure 4C), suggesting downward advective flux at P60, while relatively constant temperature (no diel variation) at similar depths suggest upward flux at P17 and P44 [Constantz, 2008]. Higher streambed temperature amplitude at P60 than equivalent depths in P44 or P17 is also observed during winter conditions, however, this was over a nearly 2 d period (Figures 4D, 4E, and 4F). The decreased diel amplitude at depth during the winter relative to the summer agrees with the near neutral flux estimates because diel temperature variations are a function of primarily streambed heat conduction and not advective transport. Typically, the higher river discharge

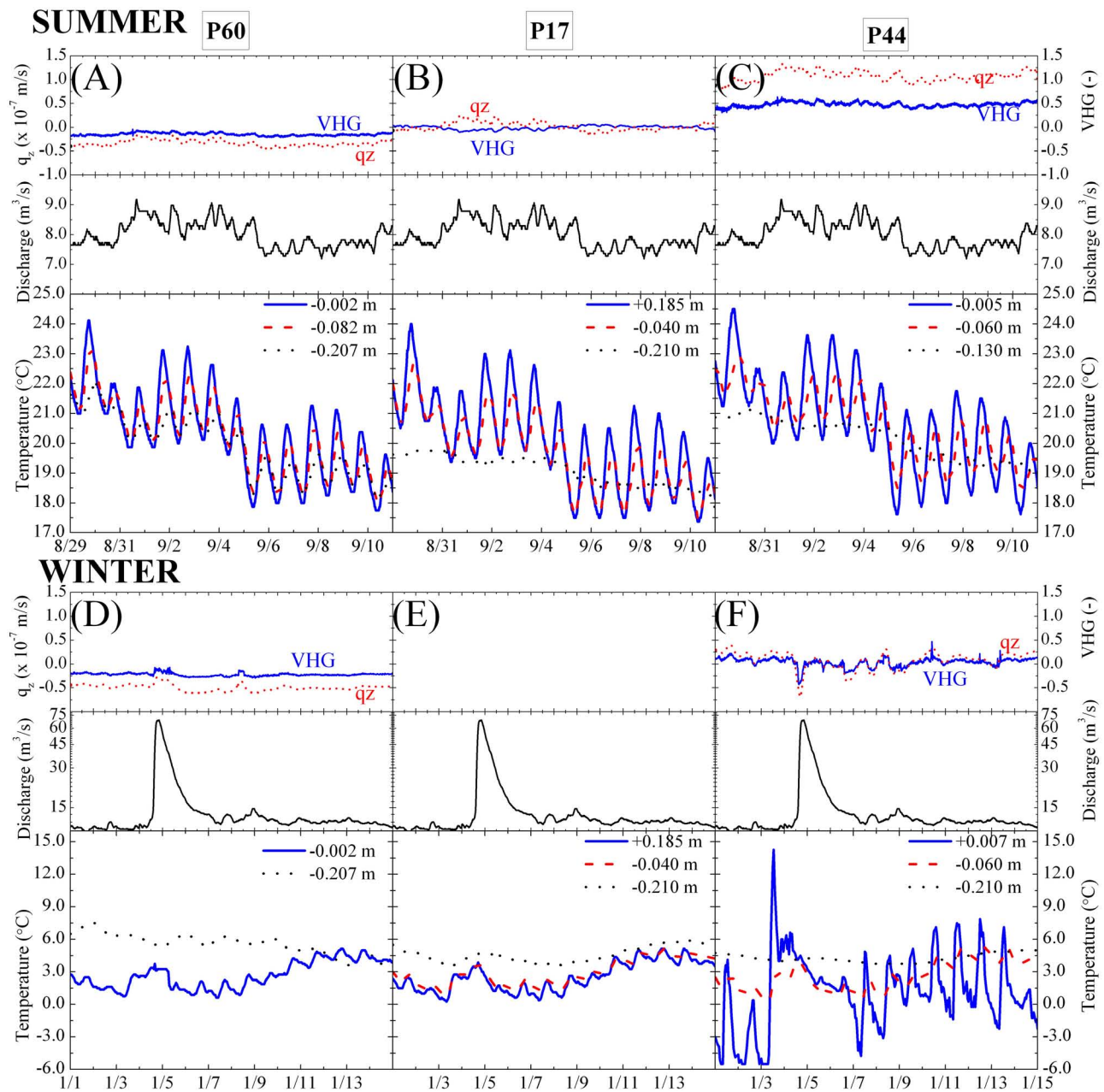


Figure 4. Summer 2007 comparison of vertical streambed flux, VHG, river discharge, and streambed thermographs at (a) the head of the MRCB (P60), (b) a midlength MRCB streambed (P17) location, and (c) near the tail of the MRCB (P44). A winter 2008 comparison is also presented for (d) the MRCB head at P60, (e) the mid MRCB at P17, and (f) the MRCB tail at P44. Note the increased temperature range during winter conditions. Positive values indicate upward flux/VHG and negative values indicate downward flux/VHG. River discharge measured at USGS streamgauge in Tracy, Nevada (10350340).

at P44 (the tail) corresponds with an increase in upward flux and an associated decrease in diel temperature change with depth (Figure 4C). During winter, near neutral vertical flux at the tail suggests that subsurface temperature response with depth is dominated by conduction rather than advection. For example, a warm precipitation event (3 January) increased river discharge and downward flux and equilibrated temperatures at depth. During hydrograph recession, the temperature amplitude with depth decreased (Figure 4F). The

flux direction qualitatively estimated from the temperature results are further supported by the observed VHG in the summer and winter.

4.2. Lateral Fluid Flux Estimates From Field Data

[26] The spatial river-channel bar fluxes are dominated by cross-bar horizontal hydraulic gradients. A drop in streambed elevation across the western channel at the MRCB head decreases the river stage by 0.2 m, which induces cross-bar

Table 3. Range in VHG and Vertical Seepage Flux (Darcy-Derived and Heat Estimated) for Streambed Locations P60, P44, and P17 Throughout Each Season

Metric	Season	Unit	P60	P44	P17
VHG	Summer	(–)	-0.133 ± 0.089	0.397 ± 0.359	0.063 ± 0.148
VHG	Autumn	(–)	-0.044 ± 0.042	0.069 ± 0.296	–
VHG	Winter	(–)	-0.247 ± 0.122	0.009 ± 0.454	–
VHG	Spring	(–)	-0.052 ± 0.048	0.427 ± 0.521	0.566 ± 0.576
q_z , Head	Summer	$(\times 10^{-7} \text{ m s}^{-1})$	-0.298 ± 0.160	0.892 ± 0.435	0.141 ± 0.217
q_z , Head	Autumn	$(\times 10^{-7} \text{ m s}^{-1})$	-0.098 ± 0.077	0.155 ± 0.478	–
q_z , Head	Winter	$(\times 10^{-7} \text{ m s}^{-1})$	-0.553 ± 0.092	0.021 ± 0.584	–
q_z , Head	Spring	$(\times 10^{-7} \text{ m s}^{-1})$	-0.054 ± 0.053	0.960 ± 1.210	1.272 ± 1.212
q_z , Thermal	Winter	$(\times 10^{-7} \text{ m s}^{-1})$	-0.481 ± 0.313	0.093 ± 1.056	0.806 ± 1.521

q_z , Head: 1-D seepage flux estimated from Darcy's law.

q_z , Thermal: 1-D seepage flux inversely estimated from optimized temperature residuals [Shope, 2009].

flow paths from the southeast head to the northwest tail of the MRCB. The stage difference between the western and eastern river channels decreases with distance downstream. As expected, transient variations in river stage lead to MRCB water table perturbations, which vary substantially between seasons (Figure 5). However, transient observed groundwater mounds and depressions develop both at seasonal and event scales.

[27] Groundwater flow paths throughout the MRCB respond quickly to stage changes, where an increase in stage typically results in an increasingly linear hydraulic gradient across the MRCB. The average transverse flux (calculated perpendicular to the spine of the MRCB) at the downstream end of the MRCB ($3.3 \pm 5.1 \times 10^{-7} \text{ m s}^{-1}$) was slightly less than the average upstream lateral flux ($4.5 \pm 4.2 \times 10^{-7} \text{ m s}^{-1}$). The net horizontal flux through

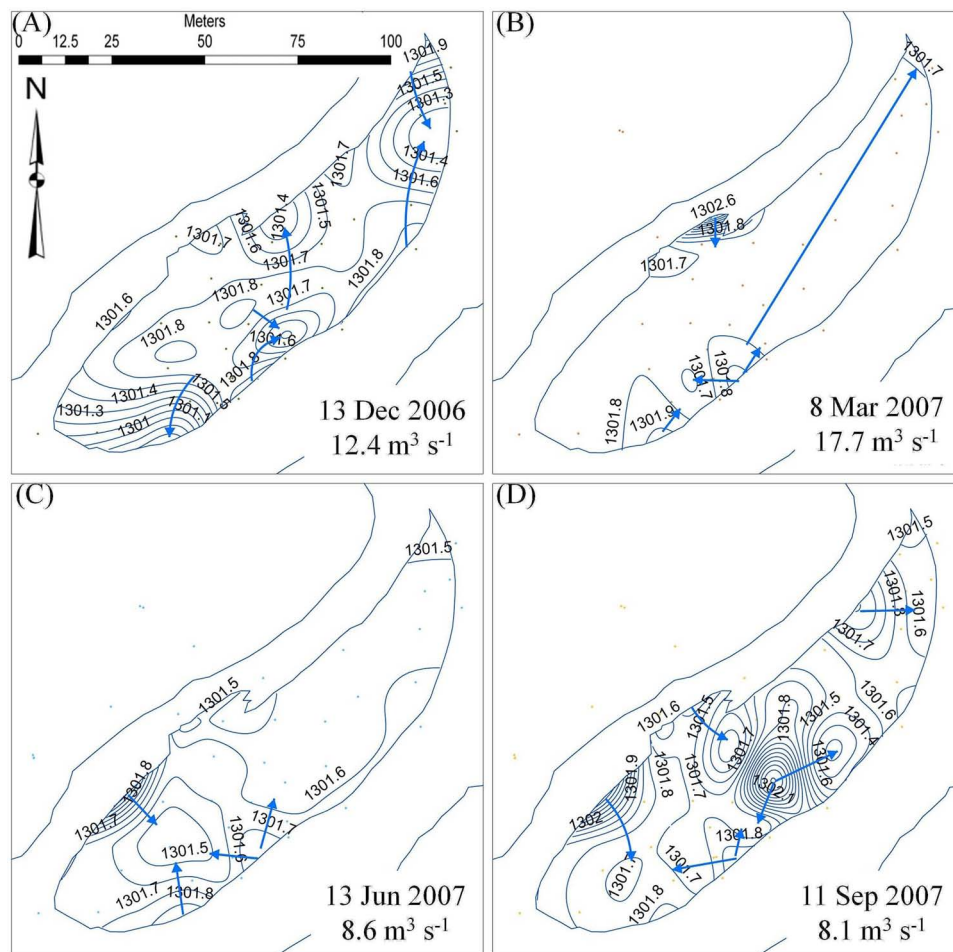


Figure 5. Channel bar potentiometric surface annotated with example transects to estimate horizontal flux. Potentiometric surfaces are shown for (a) December 2006, (b) March 2007, (c) June 2007, and (d) September 2007 to highlight the both the spatial and temporal variability in horizontal hydraulic gradients. Surface water flow is from southwest to northeast.

the middle of the MRCB was on average, an order of magnitude less than at the river-MRCB interface ($1.3 \pm 7.3 \times 10^{-6} \text{ m s}^{-1}$), particularly in the channel bar tail. As expected, the higher net flux near the river and MRCB interface was primarily due to diurnal stage variations, which typically extended up to 5–10 m into the island (Figure 6A). Alternatively, diel temperature variations were observed up to 10 m into the tail and up to 22 m at the head of the MRCB (Figure 6B).

4.3. Identification of Groundwater Mounds and Depressions

[28] On event-based and seasonal periods, the MRCB water table variations were remarkably different from the diurnal periods. Natural diel stage changes are not large enough to significantly impact the water table throughout the MRCB (Figures 7A and 7B). In winter, the hydraulic gradient changes in response to the river stage, although, the water table distribution remains consistent (Figure 7A). In summer, a long central water table ridge forms along the length of the MRCB (Figure 7B). The summer contour intervals are also 20 times greater, indicating the volume of water from the MRCB storage over the diel period.

[29] Large event-based variations in river discharge significantly changed the spatial water table patterns and MRCB drainage was dependent on the length of time of the disruption period (Figure 8). Typically, during the rising limb of an event hydrograph, the river stage increases creating a bathtub effect with flow toward a depression near the centroid of the island and near the tail of the MRCB. The water table depression typically lasts <3–4 h as the river stage increases. A cross-bar water table depression occurs immediately downstream from P29. This vegetated area (Figure 1) was formerly a flow chute, filled in during a 2006 peak flow event with medium to fine sand. Near the hydrograph peak, a relatively linear hydraulic gradient develops across the MRCB. As the river stage begins to fall, a groundwater mound at the head of the channel bar develops, similar to the results of Francis *et al.* [2010]. Further, the mound is elongated along the length of the upper half of the MRCB, forming into a ridge. As expected, this

indicates flow contribution to the depression is not only from the lateral interactions with the Truckee River but also along longitudinal flow paths through the island. The longevity of the groundwater mound is highly dependent on antecedent moisture conditions from previous events and MRCB aquifer storage. For example, the 12 April event (Figure 8A) indicates that the mound elevation decreased 6 cm in 48 h; however, the mound decreased 12 cm in 24 h on 15 May and 30 cm in 48 h after a 4 January event. In addition, the dynamics of consecutive events can be significant (Figures 8A and 8B). A groundwater mound with steep hydraulic gradients and the lack of a water table depression is observed during the first event (Figure 8A). However, the smaller event (Figure 8B) shows a shallower gradient and well-defined groundwater sink.

[30] While peak flow events are crucial in the water table distribution, seasonal trends are also important. From winter through snowmelt-dominated runoff conditions (January–May), the MRCB hydraulic gradient increases with river stage increases, and during the seasonal hydrograph recession a groundwater mound develops near the head of the island (Figure 9). Individual events contribute to the water table distribution and enhancement of mound shape and size (Figure 8). During the summer (June–July), the island hydraulic gradient is essentially flat. From late summer through the fall (August–October), the MRCB water table ridge becomes longer and the water table was consistently much higher than the river stage with steep hydraulic gradients. Throughout the winter (November–December), the water table distribution again becomes flat with a relatively consistent hydraulic gradient from the head to the tail. These results indicate that at the MRCB, long-term (seasonal) antecedent moisture and storage effects, particularly during late summer, have a significant influence on groundwater mound development and that the hydraulic conductivity distribution was not the primary influence.

4.4. Groundwater Flow and Heat Transport Simulation

[31] To support our hypothesis of complex flow patterns through the MRCB, the streambed, and the stream bank,

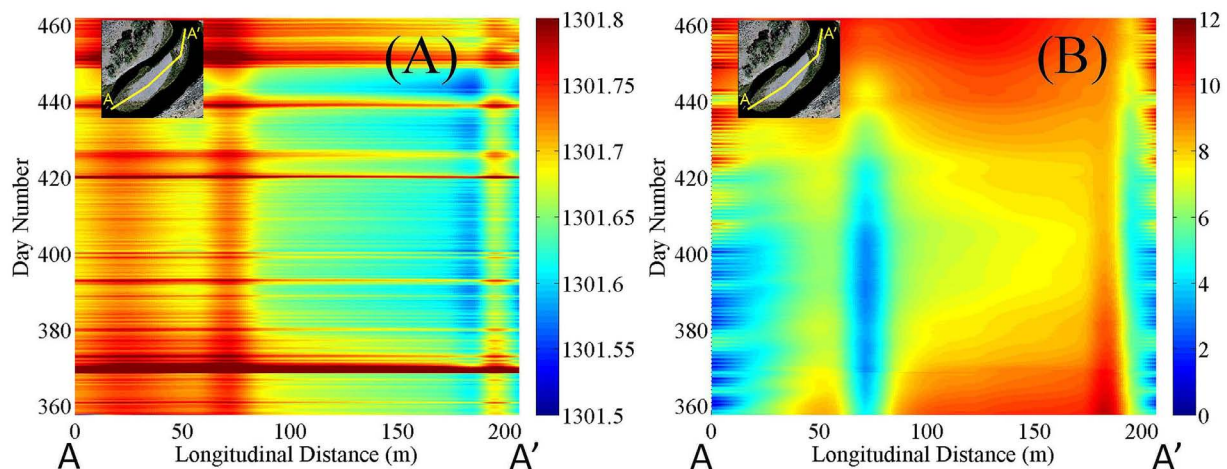


Figure 6. (a) Longitudinal MRCB water level elevation from 1 January 2007. (b) Longitudinal MRCB temperature, as inferred from river and piezometers ranging between the stream interface up to 1.2 m below ground surface (bgs).

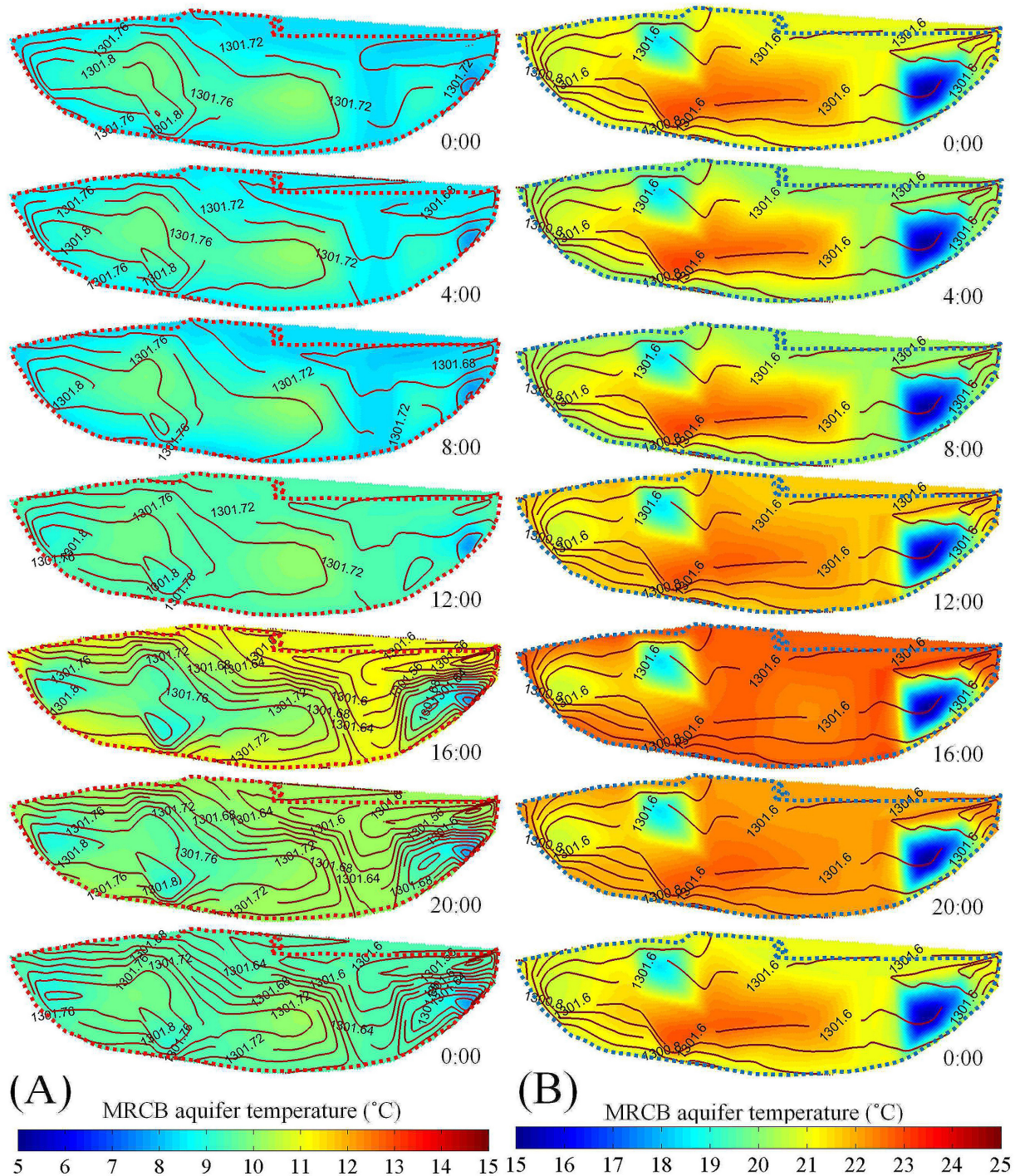


Figure 7. (a) Typical winter diel variations in MRCB water level elevation (2 cm contours) and temperature (color bar) on 12 January 2008. (b) Typical summer diel variations in MRCB water level elevation (40 cm contours) and temperature (color bar) on 28 August 2008.

we constructed a 3-D numerical flow and transport model. The model was used to estimate spatial flux variations over a short, 6 d, highly instrumented winter period to corroborate long-term field-based estimates.

[32] Based on the spatial and temporal model RMSE, the study area heterogeneity is adequately described in the 3-D flow and transport model, although smaller flow systems are embedded within the overall near-stream interactions

[Shope, 2009]. The model was calibrated to 40 hydraulic head points and 79 temperature observation points in 28 individual wells and piezometers at hourly timesteps. The maximum head residual was 0.084 m over the 4.68 m range in head, the NSE was computed as 0.918, and the model RMSE was 0.03, indicating that 95% of the data were within 3 cm of observed values (Figure 10A). MRCB temperature results were separated into the area within 5 m of

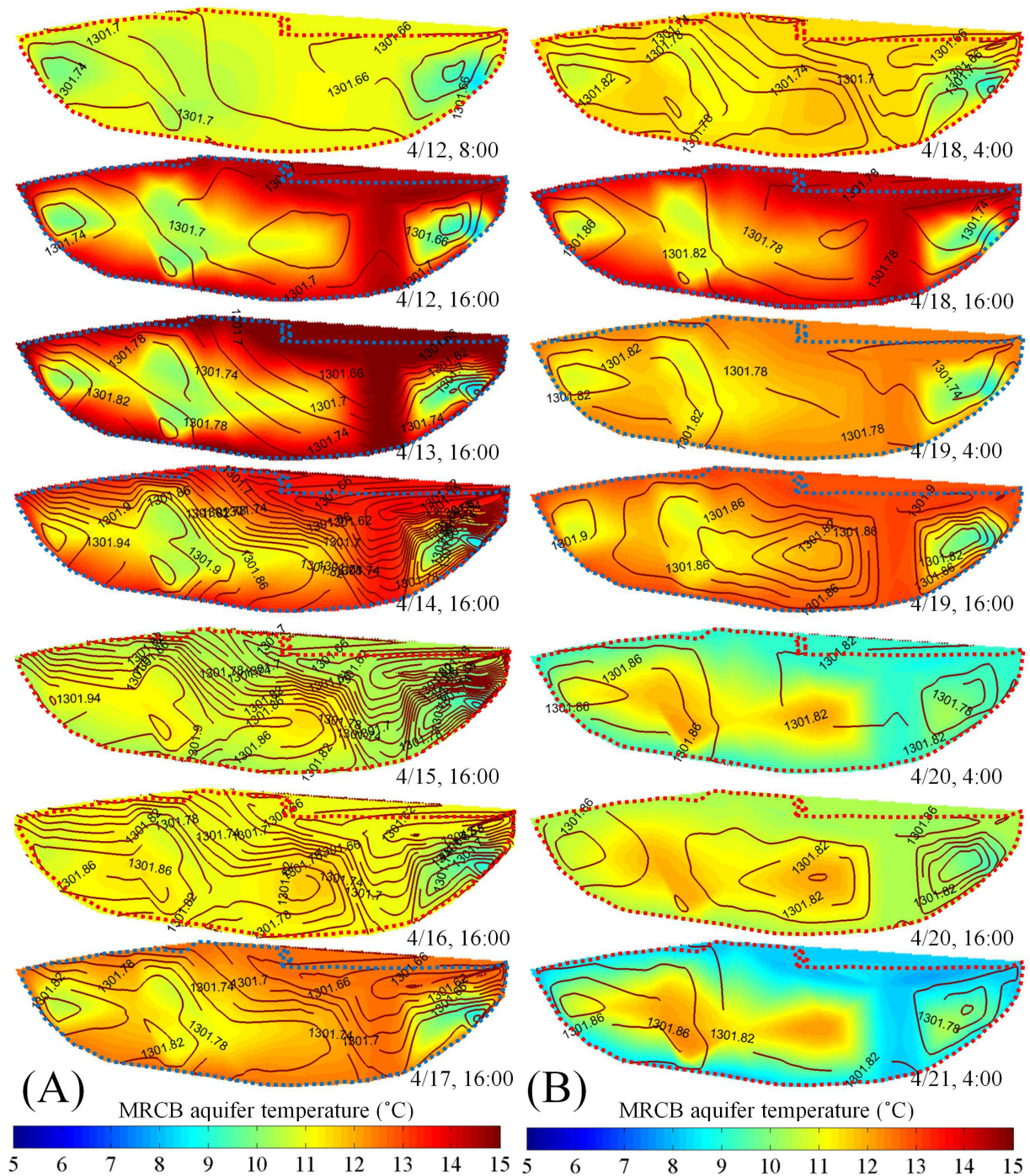


Figure 8. (a) Synopsis of peak event variations in MRCB water level elevation (2 cm contours) and temperature (color bar) between 12 April and 17 April 2008. (b) Synopsis of a second peak event immediately after the 12 April event, with variations in MRCB water level elevation (2 cm contours) and temperature (color bar) between 18 April and 21 April 2008.

the river interface and the central portion of the MRCB, which resulted in slightly better midbar (R^2 0.74) and near-stream (R^2 0.83) temperature results than the overall average (R^2 0.63) (Figure 10B). Further, hydraulic conductivity variations decreased temperature residuals in some locations but increased in other locations, suggesting that three hydraulic conductivity units may not be sufficient. Perhaps

as Cardenas [2010] suggests, an exponential decay in hydraulic conductivity with the distance from the head and the banks of the channel bar would be appropriate. However, we do not have the information to justify this and the discrete values represent the overall system heterogeneity.

[33] A (ΔT_N) metric is used to compare model temperature error between piezometers representing different

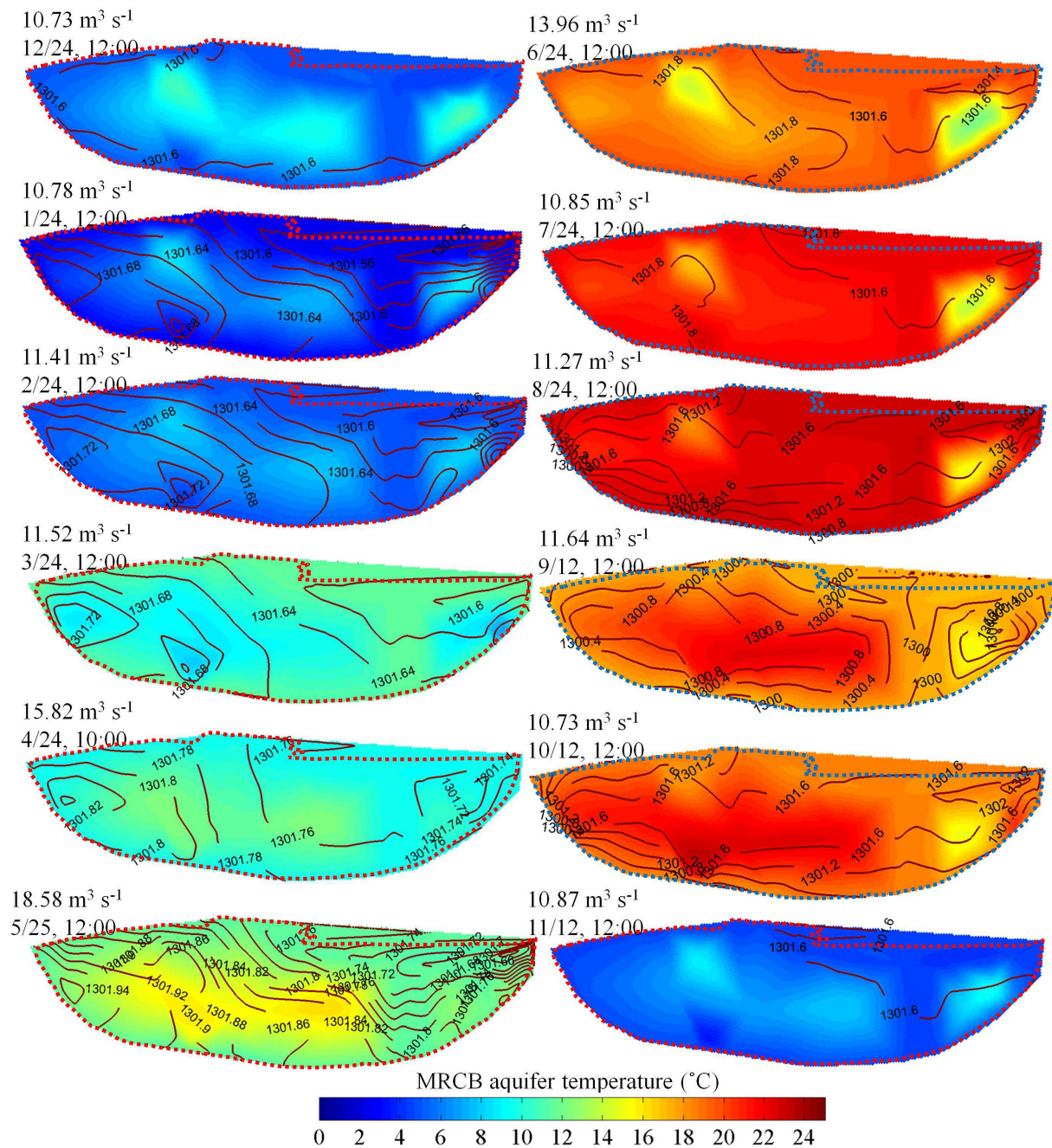


Figure 9. Seasonal water table distribution of the MRCB. Contour intervals vary by month for visualization, where: January, February, March, April, and May equal 2 cm, June, July, November, and December equal 20 cm, August, September, and October equal 40 cm. Dates and times were varied in some cases to identify seasonal trends rather than event-based water table distributions.

geomorphic locations throughout the length of the simulation. The average absolute temperature residuals over the length of the simulation period (n) at each thermistor are normalized to the observed temperature range ($*T_{\text{obs}}$) of the individual thermistor, where (ΔT_N) is calculated as,

$$\Delta T_N = \frac{\sum_{i=1}^n (T_{\text{obs}} - T_{\text{sim}})}{n} \bigg/ (*T_{\text{obs}}). \quad (3)$$

[34] The ΔT_N is near zero with minimal deviation for most MRCB locations, indicating minor residual bias over the range in observed temperatures (Figure 11). The ΔT_N for the streambed locations is also near zero; however, the increased standard deviation indicates temporal differences in residuals. The generally higher ΔT_N in the stream bank indicates increased temperature residuals relative to the minimal observed temperature range in the stream bank wells. For some stream bank locations the $*T_{\text{obs}}$ through the study period varied $<0.1^{\circ}\text{C}$, resulting in a large ΔT_N .

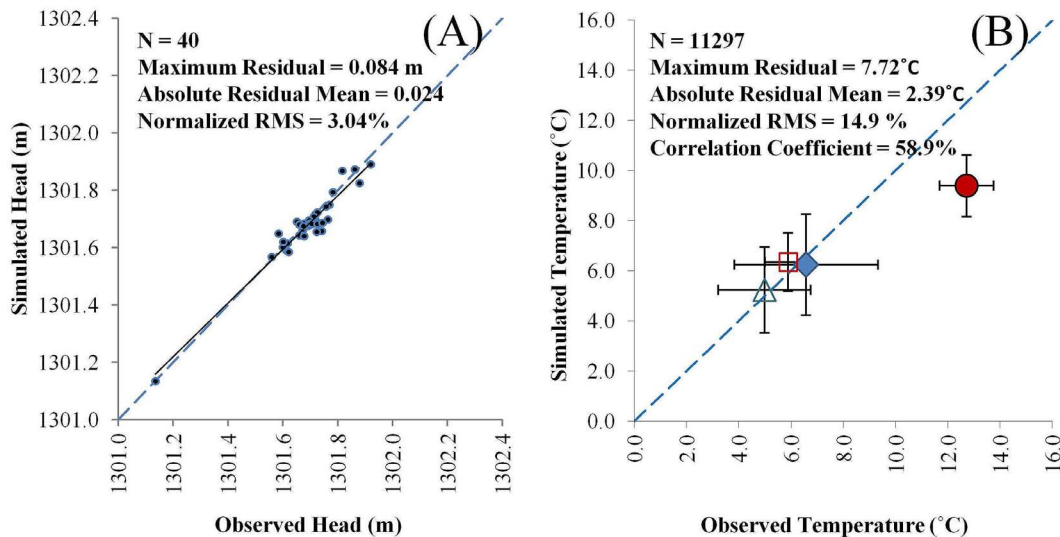


Figure 10. (a) Observed versus simulated hydraulic head values for all 40 wells and piezometers throughout the channel bar, the streambed, and the stream bank. The blue dashed line corresponds to the 1:1 relationship and the solid black line corresponds to the linear trendline of all data points. (b) Observed versus simulated temperature for 11,297 channel bar, streambed, and stream bank measurements throughout the study period. Mean temperature comparison with observed and simulated standard deviations is shown for the stream bank (red closed circle) and the streambed (blue closed diamond). The entire channel bar is discretized into the central channel bar (open square) and channel bar locations within 5 m from stream (open triangle). The blue dashed line corresponds to the 1:1 relationship.

[35] Simulated near-stream groundwater flow directions are multidirectional and spatially variable and the total near-stream flux from the numerical simulations ranged between 1.2×10^{-13} and $9.9 \times 10^{-6} \text{ m s}^{-1}$ (Figure 12). The optimized hydraulic conductivity values for the channel bar, the streambed, and the stream bank shallow subsurface were 6.5×10^{-5} , 5.7×10^{-4} , and $3.4 \times 10^{-6} \text{ m s}^{-1}$, respectively. Net flux into the MRCB dominated along the eastern length of the channel bar, although net flux along the western MRCB was spatially variable and dependent on stream geometry (Figures 12 and 13). The total net river-groundwater interactions ($\text{m}^3 \text{ d}^{-1}$) normalized to bar length (m) was $0.029 \text{ m}^3 \text{ d}^{-1} \text{ m}^{-1}$ length of stream bank. Generally, the simulated vertical streambed flux ranged

between -3.0×10^{-6} and $5.0 \times 10^{-6} \text{ m s}^{-1}$ and the MRCB lateral flux estimates ranged between 2.3×10^{-9} and $10.0 \times 10^{-6} \text{ m s}^{-1}$. While some streambed locations (P44) indicated high vertical flux (Figure 4C), the 3-D simulation results quantify two orders of magnitude more lateral than vertical flux.

[36] To further quantify the MRCB's effect on near-stream fluxes, the simulated channel bar was removed from the model and the net flux was again estimated. The normalized horizontal flux rate between the river and MRCB along the perimeter of the channel bar was summed with the total vertical flux throughout the MRCB and estimated as $6.2 \times 10^{-6} \text{ m s}^{-1}$. The total vertical flux through the adjacent bed was normalized to the MRCB cross-sectional

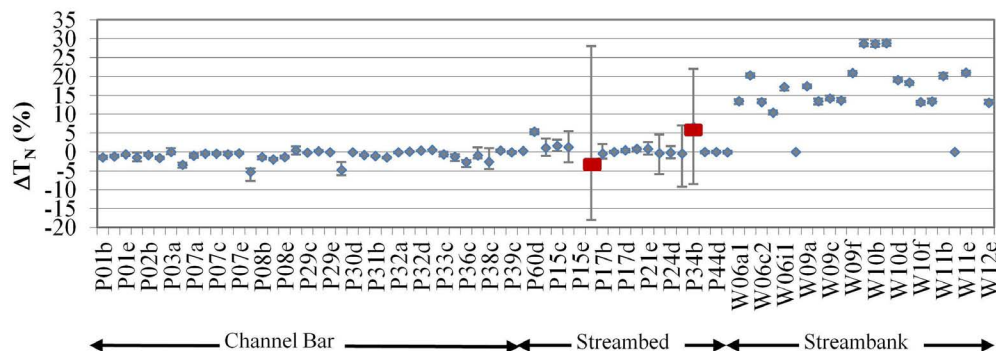


Figure 11. Normalized difference between observed and simulated temperature throughout the simulation at each thermistor. The blue diamond represents the average temperature difference over time at each location. The error bars represent the observed range in temperature. The red squares denote locations P17a and P43a, which are further discussed in the text.

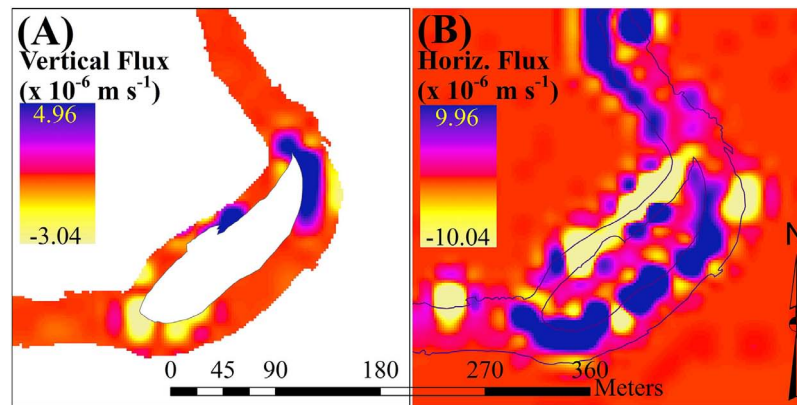


Figure 12. The 3-D flow and transport model simulated (A) vertical seepage flux ($\times 10^{-6} \text{ m s}^{-1}$) at the streambed interface where the piezometer screened depth is generally 0.4 m, and (B) lateral seepage flux ($\times 10^{-6} \text{ m s}^{-1}$) between the river and the channel bar, along flowlines within the streambed, and between the river and stream bank area. Vertical seepage flux across the water table of the stream bank and the island is not included (white) due to the prescribed no flow boundary condition.

area to estimate the net streambed flux in the study reach of $9.8 \times 10^{-8} \text{ m s}^{-1}$. Therefore, the total flux rate calculated throughout the channel bar was 6.4 times greater than the equivalent vertical streambed flux alone. Similar results were found during a solute tracer modeling study at an adjacent upstream reach following an extensive restoration project [Knust and Warwick, 2009]. Our results suggest that dynamic channel bars may provide similar river-groundwater interactions as those induced through restoration processes.

5. Discussion

5.1. Spatial Flux Variability Between Geomorphic Units

[37] The relative flux contributions between the channel bar, the streambed, and the stream bank were substantially different, suggesting that care be taken in the classification of specific near-stream geomorphic units. Event-based aggradation and sorting followed by periodic sediment

redistribution and mass movement create highly heterogeneous layered structure; therefore, different morphological features are expected to exhibit different influences on river-aquifer processes. Overall, measured hydraulic conductivity estimates ranged from 3.3×10^{-7} to $2.0 \times 10^{-4} \text{ m s}^{-1}$. The 1-D vertical streambed hydraulic conductivity used in Darcy calculations was $2.2 \times 10^{-6} \text{ m s}^{-1}$, consistent with the range in temperature-optimized estimates (1.3 to $4.9 \times 10^{-6} \text{ m s}^{-1}$). Further, optimized hydraulic conductivity estimates from the 3-D numerical model indicated that the streambed hydraulic conductivity was an order of magnitude higher than the MRCB, and 2.5 orders of magnitude higher than the adjacent stream bank.

[38] The streambed was typically characterized by higher lateral than vertical flux and attributed to significant streambed sediment anisotropy. Lateral streambed fluxes were typically 2–5 times higher than the vertical flux, although in some locations during specific times of the year (i.e., winter at P44) the lateral flux was more than an order of magnitude higher. A patchwork of vertically dominated streambed flux was observed, and is expected based on streambed morphology flow patterns [Elliott and Brooks, 1997a, 1997b; Thibodeaux and Boyle, 1987] and in-stream channel geometry [Revelli et al., 2008; Woessner, 2000].

[39] Overall, fluxes were higher from the streambed than the MRCB and the MRCB fluxes were higher than the stream bank contributions. The MRCB and stream bank domain were dominated by lateral flux (Figure 12B) with flow typically focused perpendicular to the river. Stage differences between the western and eastern river channel of the MRCB induce cross-bar flow patterns from the east to the west bank, which coupled with increased hydraulic conductivity result in typical flux values 1.5 orders of magnitude greater than the stream bank. Obviously, streambed topography and in-stream channel geometry increase the MRCB interface flux variability, as shown in Figures 12 and 13. Nevertheless, higher lateral flux was generally estimated at the eastern portion of the MRCB head and tail and exfiltration increased with distance downstream on the western MRCB bank.

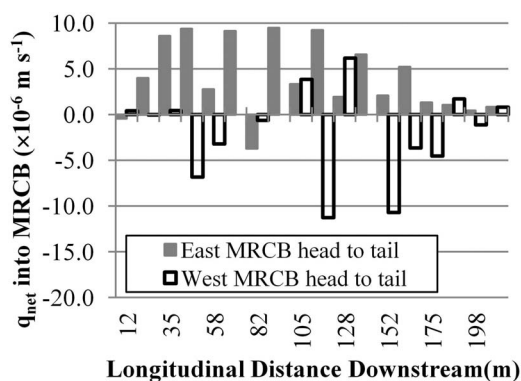


Figure 13. Net flux into the channel bar plotted versus the longitudinal distance downstream. Positive flux rates indicate that surface water is infiltrating and the MRCB is gaining water from the stream and negative flux rates indicate exfiltration from the MRCB. Refer to Figure 1 for the MRCB western and eastern transect locations.

5.2. Impacts of Nonvertical Flow

[40] Vertical head gradients are typically monitored along a limited number of 1-D vertical transects every several kilometers [Arntzen *et al.*, 2006; Hanrahan, 2008]. Similarly, we used both head-based and heat-based methods to calculate the vertical 1-D streambed flux at 11 locations and our results showed a reasonable comparison. However, when the 1-D estimates were compared to multi-dimensional results, the strictly vertical flow assumption was invalidated at many locations, particularly in response to an event or seasonal discharge variations. For example, our 3-D model results indicated that the streambed vertical flux component at the MRCB tail (P44) was two orders of magnitude less than the horizontal flux during the winter, although our 1-D analysis was within reasonable flux range and corroborated by multiple methods. Wondzell and Swanson [1999] also found that significant vertical flux was not observed in streambed wells, but that flow contributions dominated from the stream banks along the lower portions of gravel bars. Further, the specific location within the channel cross-section can influence river-groundwater interactions. Shanafield *et al.* [2010] found that lateral fluxes were over 8 times higher near the stream banks than at the center of the channel. Alternatively, we observed that upward flux dominated at P17 throughout most seasons, and was corroborated with the 3-D numerical simulations (Table 3 and Figure 3). Because of the complexity and more extensive data requirements of a 3-D model, their use has been fairly limited in heat tracer studies. However, surface water and groundwater interactions have often been shown to be multidimensional and multiscale [Cardenas, 2008; Poole *et al.*, 2006; Storey *et al.*, 2003]. Further, previous studies discuss hyporheic flow patterns in large fluvial islands under natural [Dent *et al.*, 2007] or dam-influenced [Francis *et al.*, 2010] conditions; however, limited studies have characterized the transient multidimensional head gradients that affect river and groundwater interactions.

[41] Complex 3-D flow patterns can substantially violate the 1-D vertical assumption at many streambed locations, but there is also a strong transient link to bank storage and antecedent moisture conditions. Several studies [i.e., Shanafield *et al.*, 2010; Vogt *et al.*, 2010] have described the difficulty in matching 1-D temperatures and suggested that it may be a result of multidimensional fluxes. In fact, Lautz [2010] points out that the greatest source of error in analytical solutions to the 1-D heat transport equation is nonvertical streambed flow, which must surely occur as flow paths are varied in response to transient stage-induced hydraulic gradient changes. We observed that variations in flow paths develop due to bank storage and antecedent moisture conditions that are not necessarily repeatable, even under similar river discharge. Transient flow patterns were not typically significant on a diel timescale, although they were substantial on an event-based and seasonal timescale. These results complement the results of Shanafield *et al.* [2010] and Storey *et al.* [2003], where the authors demonstrated that as the hydraulic gradient increases, total infiltration increases, and the vertically dominated flux distribution shifts to laterally dominated. Our results further elucidate the transient antecedent soil moisture and bank storage conditions responsible for the seasonality. In particular, Anibas *et al.* [2009] demonstrated that vertical flux can be

approximated with a steady state analytical solution for specific times of the year (summer and winter). It stands to reason that while 1-D vertical flux estimates may be advantageous in certain applications, without prior knowledge of both the spatial flow patterns and the transient head distribution, results become increasingly uncertain.

5.3. Flux Reversals and Transition Periods

[42] Similar to results from Barlow and Coupe [2009], the vertical streambed seepage direction at many locations varies not only by event but also seasonally. For example, at location P17, a nonlinear stage threshold defines the flux direction reversals on a daily to event-based period [Shope, 2009]. The observed flux direction reversal may be attributed to the unique balance between vertical stage-dependant and lateral island storage-dependant hydraulic gradients. Similar to results presented by Fanelli and Lautz [2008], vertical flux direction reversed at specific locations while adjacent locations showed no directional flux change, which is likely attributed to the complex lateral flow path contributions. With low hydraulic head gradients, small stage variations have relatively large impacts on the VHG and flux estimates and each location has transitional periods when the hydraulic head difference is minimized (Figure 3F). Streambed transition periods were typically during the early spring and fall (i.e., MRCB head), although the transition period at the tail was during the fall and winter (Table 3). The rapid flux reversal at P17 suggests that contributing flow paths are relatively short, although the seasonal flux reversal at P44 suggests longer lateral flow paths draining the island.

5.4. Groundwater Mound Development

[43] The general development of the fluvial island water table features are similar to previous studies; however, the MRCB mound observations are unique because they show transient behavior at different observational timescales, they vary in spatial extent, and most importantly, are not necessarily repeatable. Our results complement those of Francis *et al.* [2010] and Cardenas [2010], where diel dam-influenced stage variations created a diurnally repeatable groundwater mound and sink in a Texas channel bar. Our results indicate that for the MRCB aquifer properties, diel variations to the MRCB water table distribution are not significant. Average diel head variations were 2–3 cm with peak variations on the order of 18–20 cm d⁻¹. Although, the diel influence of stream stage and temperature are limited to ~20 m around the river-channel bar interface.

[44] However, on an event basis, water table variations are more significant, with mound and sink development, consistent with the observations of Francis *et al.* [2010] and Cardenas [2010]. River stage changes are rapidly propagated throughout the entire fluvial island. As the MRCB water table increases and subsequently begins to decline, groundwater mound features typically develop at the head of MRCB and a downstream mound was occasionally located at the tail (Figures 7, 8, and 9). A groundwater sink was occasionally present between the upstream groundwater mound and the flow chute, and typically corresponding to the rising limb of the hydrograph. The unsaturated soil conditions were not monitored throughout this study, and therefore it is impossible to quantitatively assess the soil moisture contributions. However, observations show that when surface water discharge

variations of similar magnitude occur within ~ 8.5 d, the typical groundwater mound and water table distribution are altered. For example, both the water table variations and the transient storage are different between consecutive events on 12 April and 18 April (Figure 8).

[45] On a seasonal basis, the water table gradient is relatively low during the winter and spring when event-based and spring runoff discharge changes can be substantial. During the summer and fall periods, the MRCB shows a high water table ridge that runs the length of the fluvial island with steep hydraulic gradients around the perimeter of the channel bar (Figure 9). These steep head gradients obviously impact the flow paths that contribute to the adjacent streambed piezometer locations. Of course, there is a dependency on the magnitude of river discharge change, precipitation events, the time of year, but what these results show is that the spatial and temporal variability suggests that antecedent moisture conditions and fluvial island storage play an important role in the overall flow patterns.

[46] This means that while the influence of groundwater mounds and sinks on a single diel cycle can be used to estimate the potential impact on local nutrient attenuation and near-stream biogeochemical transformations, the implications on the reach-scale have a strict dependency on longer-term and time-specific antecedent moisture and storage responses. For example, using the estimate of island storage exchange provided by *Francis et al.* [2010] and the mean daily river discharge, the net island storage contribution to the river discharge is 0.13%. Therefore, assessment over a single day or even an event may be inadequate to characterize the impact of river-aquifer interactions in a natural fluvial island. The combination of relative stage change, the length of time that the stage remains high, the moisture content, and transient 3-D flow patterns with higher residence time contribute to longer-term fluvial island impacts. As mentioned, the variably saturated soil moisture conditions were not considered in this study, but in conjunction with transient estimates of island storage would provide for an interesting study based on these results.

[47] The spatial and temporal water table variability may be attributed to a number of hydrologic processes. The spatial water table distribution suggests that the groundwater features are a function of increased sediment heterogeneity or subsurface velocity. Since increased river stage typically decreases the hydraulic gradient across the MRCB potentiometric surface, the mound and sink development over time suggests that channel bar heterogeneity may have an influence. As described by *Cardenas* [2010], fluvial bars typically exhibit downstream grain size fining. While with the aquifer properties observed at the MRCB, the specific yield decreases with decreasing grain size, and the specific retention increases. This suggests that groundwater mounds would have a higher probability of occurrence at downstream locations characterized by increased fine sediment. We have also considered the possibility that sediment deposition at the streambed or during periodic island inundation may contribute to clogging, and subsequently effect transient mound development and possibly feature location. *Nowinski et al.* [2011] found that spatiotemporal permeability variations occur at a similar bar scale. However, because the channel bar is inundated on average every 2.5 yr, the spatial variability of the groundwater mounds

within a seasonal response time does not agree. The annual variability is also not likely a function of streambed clogging because similar responses are observed during similar seasonal periods every year. This suggests that the water table response is a function of hydrograph-influenced variability of MRCB storage and antecedent moisture conditions, typically repeated on an annual basis.

6. Summary and Conclusions

[48] A spatially dense vertically and horizontally distributed array of field monitoring locations was used in conjunction with 3-D numerical simulations to examine near-stream interactions between a channel bar, streambed, and stream bank over several years. Because of the difficulty involved with instrument installation in gravel bar and cobble streambeds and the complexity of multidimensional numerical modeling, similar studies are rare. To our knowledge, the 3-D modeling approach described in this study is the first to examine field conditions and quantify flow and transport in and around a channel bar.

[49] Our results show that channel bars significantly impact multidimensional fluxes and based on a multiyear record of field data, that this impact substantially varies based on season and event-based conditions. Long-term flow responses to event-based and seasonal geomorphic changes have been examined [*Kasahara and Wondzell*, 2003; *Wondzell and Swanson*, 1996; *Wondzell and Swanson*, 1999; *Wroblicky et al.*, 1998]. However, a combination of both individual event-based conditions and the near-stream response coupled with seasonal investigations has been limited. *Francis et al.* [2010] examined temporal conditions of a channel bar and showed with 2-D planimetric analysis that channel bar water levels do vary over a diurnal period during base flow conditions. *Cardenas* [2010] concluded from a modeling study that while useful for qualitative similarities to field results, process-based temporal and spatial channel bar dynamics could not be fully explained by a Boussinesq-type model. Our study builds on the knowledge gained from these important channel bar studies by examining the long-term spatial and temporal dynamics and quantitatively assessing spatial process-based interactions with a 3-D numerical model. The 3-D simulation results and long-term field analysis indicate that flux estimates of the channel bar vary significantly from the streambed and stream bank. We describe the interactions of multiple influences from controls such as streambed topography, channel meanders, and hydraulic conductivity. In fact, the MRCB was found to exhibit over 6 times more flux than a similar stream without the channel bar. This increase in near-stream flux due to the bar suggests the possibility that restoration activities including channel bar implementation can increase near-stream interactions. These results were consistent with a study near the same location by *Knust and Warwick* [2009], where solute tracers indicated that near stream interactions increased nearly 6 times at recently restored reaches. The study results clearly indicated in a quantitative manner the importance of stream interactions with channel bars relative to other near-stream sediments, and that channel bars increase hyporheic flow.

[50] These relative magnitudes of diurnal and event-based flux rates and their spatial extent suggest that a

similar trend should exist for biogeochemical reactions, which have been shown to be strongly coupled to flow and transport processes (i.e., organic rich gravel bars may act as sinks [Pinay et al., 1994] or sources [Triska et al., 1993] of inorganic nutrients). Near-stream interactions can also moderate diel temperature range in surface and subsurface environments and create habitat for fish and macroinvertebrates [Arrigoni et al., 2008]. Increasing the river channel complexity through fluvio-geomorphic features such as channel bars, can add ecological functioning by increasing the near-stream interactions [Fernald et al., 2006]. Kasa-hara and Hill [2007] and Dent et al. [2001] provide evidence that gravel bar construction or cumulative channel bars can increase flow interactions and the potential for nutrient cycling. Our study provides an in-depth spatial and temporal data set and modeling of channel bar interactions. While our study used temperature variations to estimate subsurface transport, in the future, other nonconservative tracers, as well as conservative tracers, may increase the spatial scale for further examination of the flux magnitude beneath the channel bar relative to the streambed and stream banks for both semiarid and other hydrologic settings.

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